Lecture 6, Jan 05, 2025.

Rates of change. Suppose y is a quantity that depends on another quantity x, $y = f(x)$. If x changes from x_1 to x_2, then the change in x is

$$\Delta x = x_2 - x_1$$

and the corresponding chang in y is

$$\Delta y = f(x_2) - f(x_1)$$

the difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_2}$$

is called the average rate of change of y with respect to x over the interval $[x_1, x_2]$.

Def. Instantaneous rate of change $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Example: (for you). A manufacturer produces bolt of a fabric with a fixed width. The cost of producing x yards of this fabric is $C = f(x)$ dollars.

a). what is the meaning of the derivative $f'(x)$? what are its units ?

b). In practical terms, what does it mean to say that $f'(1000) = 9$?

c). which do you think is greater $f'(50)$ or $f'(500)$?
what about $f'(5000)$?

2.2. The Derivative as a function.

We know that

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Let $a = x$ then

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Given any number x for which this limit exists, we assign to x the number $f'(x)$, f' is a new function called the derivative of f

Example. a). if $f(x) = x^3 - x$ find a formula for $f'(x)$.

b). Illustrate this formula by comparing the graphs of f and f'

Sln a).

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

$f(x+h) = (x+h)^3 - (x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - x - h$

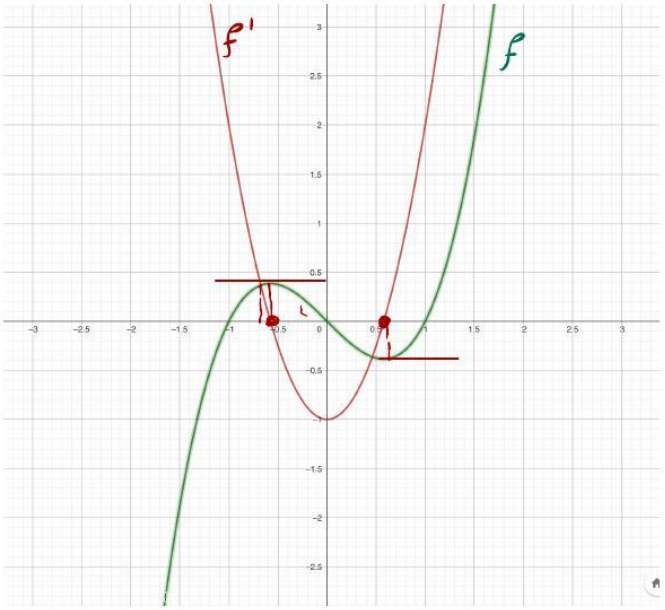
$f(x) = x^3 - x$

$$f'(x) = \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x - h - x^3 + x}{h}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$$

$$= \lim_{h \to 0} \frac{h(3x^2 + 3xh + h^2 - 1)}{h} = 3x^2 - 1$$

b).



Notice that $f'(a)=0$ then f has horizontal tangents. and $f'(x)\geq 0$ when the tangents has positive slope and $f'(x)\leq 0$ when the tangents has negative slopes.

Example. If $f(x)=\sqrt{x}$ find the derivative of f. State the domain of f'.

Sln.

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}\times\frac{(\sqrt{x+h}+\sqrt{x})}{(\sqrt{x+h}+\sqrt{x})}$$

$$=\lim_{h\to 0}\frac{(\sqrt{x+h})^2-(\sqrt{x})^2}{h(\sqrt{x+h}+\sqrt{x})}$$

$$=\lim_{h\to 0}\frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}$$

$$=\lim_{h\to 0}\frac{1}{\sqrt{x+h}+\sqrt{x}}=\frac{1}{2\sqrt{x}}$$

then $\text{Dom } f'=(0,\infty)$ and $\text{Dom } f=[0,\infty)$.

Problem Find f' if $f(x)=\dfrac{1-x}{2+x}$

other Notations.

$$f'(x)=y'=\frac{dy}{dx}=\frac{df}{dx}=\frac{d}{dx}(f(x))=Df(x)=D_x f(x).$$

the symbols D and $\frac{dy}{dx}$ are called differentiation operators.

If we want to indicate the value of $\frac{dy}{dx}$ in the Leibniz notation at a particular number a.

$$\left.\frac{dy}{dx}\right|_{x=a} \quad \text{or} \quad \left.\frac{dy}{dx}\right]_{x=a} \quad \text{or} \quad f'(a).$$

Def. A function f is **differentiable at a** if $f'(a)$ exists. It is differentiable on an open interval (a,b) or (a,∞) or $(-\infty,a)$ or $(-\infty,\infty)$ if it's differentiable at every number in the interval.

Example. where is the function $f(x)=|x|$ differentiable?

Sln. we know

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

if $x>0$ then $|x|=x$ then

$$f'(x) = \lim_{h\to 0} \frac{|x+h|-|x|}{h} = \lim_{h\to 0} \frac{x+h-x}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

So, f is differentiable for every $x>0$.

if $x<0$: we have $|x|=-x$ So $x+h<0$

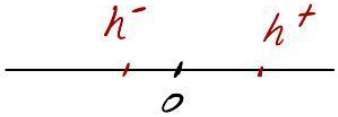
$$f'(x) = \lim_{h\to 0} \frac{|x+h|-|x|}{h} = \lim_{h\to 0} \frac{-(x+h)-(-x)}{h}$$

$$= \lim_{h\to 0} \frac{-x-h+x}{h} = \lim_{h\to 0} \frac{-h}{h} = -1.$$

then f is differentiable for every $x<0$

for $x=0$ then

$$f'(0) = \lim_{h\to 0} \frac{|0+h|-|0|}{h} = \lim_{h\to 0} \frac{|h|}{h}$$



$$\lim_{h\to 0^+} \frac{|h|}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

$$\lim_{h\to 0^-} \frac{|h|}{h} = \lim_{h\to 0} \frac{-h}{h} = -1.$$

thus $f'(0)$ does not exists. So f is not differentiable at $x=0$.

$$f'(x) = \begin{cases} -1 & \text{if } x<0 \\ 1 & \text{if } x>0. \end{cases}$$

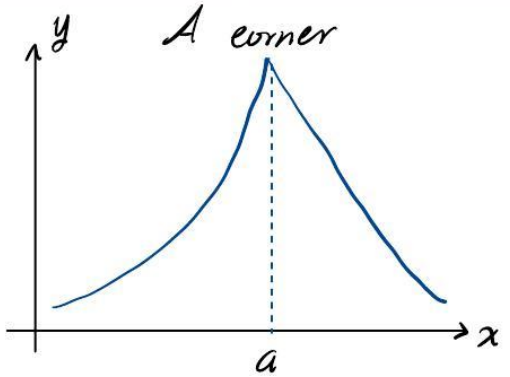
Thm. If f is differentiable at a then f is continuous at a.

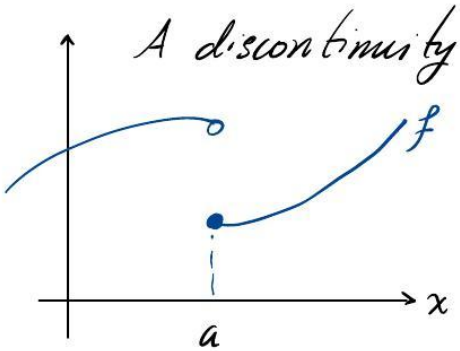
The converse is not true.

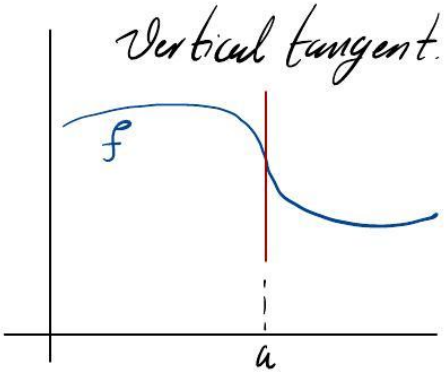
⚠ if f is continuous at a then f is differentiable at a.

Example. $f(x)=|x|$ is continuous at $x=0$ but it's not differentiable at $x=0$.

Cases where $f'(a)$ does not exists.







Higher Derivatives.

If f is a differentiable function, then, it's derivative f' is also a function, so f' may have derivative of its own, denoted by $(f')' = f''$ this function is called the second derivative of f.

$$\underbrace{\frac{d}{dx}}_{\text{derivative of}}\underbrace{\left(\frac{dy}{dx}\right)}_{\text{first derivative}} = \underbrace{\frac{d^2y}{dx^2}}_{\text{second derivative.}}$$

Example. if $f(x) = x^3 - x$ find $f''(x)$.

Sln. We know that $f'(x) = 3x^2 - 1$

$$f''(x) = (f')'(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

$$= \lim_{h \to 0} \frac{(3(x+h)^2 - 1) - (3x^2 - 1)}{h}$$

$$= \lim_{h \to 0} \frac{(3(x^2 + 2xh + h^2) - 1) - 3x^2 + 1}{h}$$

$$= \lim_{h \to 0} \frac{3x^2 + 6xh + 3h^2 - 1 - 3x^2 + 1}{h}$$

$$= \lim_{h \to 0} \frac{6xh + 3h^2}{h}$$

$$= \lim_{h \to 0} \frac{h(6x + 3h)}{h} = 6x$$

The third derivative f''' is the derivative of f''

$$y''' = f'''(x) = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^2}.$$

Check that $f'''(x) = 6$.

2.3. Differentiation formulas.

* **Constant functions.** Let $f(x) = c$ where $c \in \mathbb{R}$ is in general a constant function.

Derivative of a constant function

$$\frac{d}{dx}(c) = \frac{d}{dx}(f(x)) = f'(x) = 0$$

Pf. $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{c - c}{h} = \lim_{h \to 0} \frac{0}{h} = 0.$

* **Power functions.** Let $f(x) = x^n$ for $n \in \mathbb{N}$.

The Power rule: if n is a natural number.

$$\frac{d(x^n)}{dx} = \frac{d}{dx}(f(x)) = f'(x) = n x^{n-1}.$$

PF: we know that

$$(x^n - a^n) = (x-a)(x^{n-1} + x^{n-2}a + \cdots + xa^{n-2} + a^{n-1}).$$

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{x \to a} \frac{x^n - a^n}{x - a}$$

$$= \lim_{x \to a} \frac{(x-a)(x^{n-1} + x^{n-2}a + \cdots + xa^{n-2} + a^{n-1})}{(x-a)}$$

$$= a^{n-1} + a^{n-1} + \cdots + a^{n-1}$$

$$= n a^{n-1}$$

So, $f'(x) = n x^{n-1}$.

Example.

- $f(x) = x^6 \qquad f'(x) = 6x^{6-1} = 6x^5$
- $f(x) = x^{1000} \qquad f'(x) = 1000 x^{999}$.

New Derivatives from old.

The constant Multiple rule: If c is a constant and f is differentiable then $\frac{d}{dx}[c f(x)] = c \cdot \frac{d}{dx}(f(x))$.

Example. Compute f' where $f(x) = \frac{13}{7} x^{15}$.

Sln: $f'(x) = \frac{d}{dx}\left(\underbrace{\frac{13}{7}}_{c} x^{15}\right) = \frac{13}{7} \underbrace{\frac{d}{dx}(x^{15})}_{\text{power rule.}} = \frac{13}{7} \cdot 15 x^{14} = \frac{195}{7} x^{14}$.

Sum of functions. Let f and g be a differentiable functions.

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}(f(x)) \pm \frac{d}{dx}(g(x)) = f'(x) \pm g'(x)$$

Example. Compute $\frac{d}{dx}(x^8+12x^5-4x^4+10x^3-6x+5)$.

Sln. $\frac{d}{dx}(x^8+12x^5-4x^4+10x^3-6x+5) = \frac{d}{dx}(x^8)+\frac{d}{dx}(12x^5)-\frac{d}{dx}(4x^4)$

$$+\frac{d}{dx}(10x^3)-\frac{d}{dx}(6x)+\underbrace{\frac{d}{dx}(5)}_{0}$$

$$= 8x^7+12\frac{d}{dx}(x^5)-4\frac{d}{dx}(x^4)+10\frac{d}{dx}(x^3)-6\underbrace{\frac{d}{dx}(x)}_{1}$$

$$= 8x^7+12(5x^4)-4(4x^3)+10(3x^2)-6$$

$$= 8x^7+60x^4-16x^3+30x^2-6.$$

Example. Find the points on the curve $y=x^4-6x^2+4$ where the tangent line is horizontal.

Sln. Horizontal tangents occur where $y'=0$.

$$y' = \frac{dy}{dx} = \frac{d}{dx}(x^4)-6\frac{d}{dx}(x^2)+\frac{d}{dx}(4)$$

$$= 4x^3-12x$$

then $4x^3-12x=0$

$4x(x^2-3)=0$

$4x(x-\sqrt{3})(x+\sqrt{3})=0$

then at $x=0,\ x=\sqrt{3},\ x=-\sqrt{3}$

if $x=0$ $f(0)=4 \Rightarrow (0,4)$

if $x=\sqrt{3}$ $f(\sqrt{3})=-5 \Rightarrow (\sqrt{3},-5)$

if $x=-\sqrt{3}$ $f(-\sqrt{3})=-5 \Rightarrow (-\sqrt{3},-5)$

} here we have horizontal tangents.

The product Rule. if f and g are differentiable, then

$$\frac{d}{dx}(f\cdot g) = f'(x)\cdot g(x)+f(x)\cdot g'(x).$$

Rmk. $(f\cdot g)' \neq f'\cdot g'$ ∇

Example. Find $F'(x)$ if $F(x)=(6x^3)(7x^4)$

Sln. Notice that $F(x)=f(x)\cdot g(x)$ where $f(x)=6x^3$, $g(x)=7x^4$.

Using the product rule.

$$F'(x)= f'(x)\cdot g(x) + f(x)\cdot g'(x)$$

$$\begin{cases} f'(x)=18x^2 \\ g(x)=28x^3 \end{cases}$$

$$= 18x^2(7x^4) + (6x^3)(28x^3)$$

$$= 168x^6 + 126x^6 = 294x^6$$

Example. If $h(x)=x\cdot g(x)$ and it is know $g(3)=5$, $g'(3)=2$ then compute $h'(3)$.

Sln. Using the product rule.

$$h'(x)=\frac{d}{dx}(x)\cdot g(x) + x\cdot g'(x)$$

$$h'(x)=g(x)+x\,g'(x).$$

then

$$h'(3)=g(3)+3g'(3)$$

$$h'(3)=5+3(2)=11$$

The Quotient Rule. If f and g are differentiable functions

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{f'(x)\cdot g(x)-f(x)\cdot g'(x)}{(g(x))^2}$$

Example. Compute y' where $y=\frac{x^2+x-2}{x^3+6}$

Sln. $y=\frac{f(x)}{g(x)}$; $f(x)=x^2+x-2$, $g(x)=x^3+6$

$f'(x)=2x+1$ $\quad$ $g'(x)=3x^2$.

by the Quotient rule

$$y' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} = \frac{(2x+1)(x^3+6) - (x^2+x-2)(3x^2)}{(x^3+6)^2}$$

$$= \frac{2x^4 + 12x + x^3 + 6 - 3x^4 - 3x^3 + 6x^2}{(x^3+6)^2}$$

$$= \frac{-x^4 - 2x^3 + 6x^2 + 12x + 6}{(x^3+6)^2}.$$

Rmk. Don't use the quotient rule every time you see a quotient. for example Let

$$F(x) = \frac{3x^2 + 2\sqrt{x}}{x}$$

Using algebra we know $\left(\frac{x+y}{a} = \frac{x}{a} + \frac{y}{a}\right)$ then,

$$F(x) = \frac{3x^2}{x} + \frac{2\sqrt{x}}{x} = 3x + 2x^{-1/2}.$$

Rmk. if $n \in \mathbb{R}$ then

$$\frac{d}{dx}(x^n) = n x^{n-1}$$

Example. if $f(x) = x^{\pi}$ then $f'(x) = \pi x^{\pi - 1}$

- Let $y = \frac{1}{\sqrt[3]{x^2}}$ we know $\sqrt[n]{x} = x^{1/n}$ $x^{-n} = \frac{1}{x^n}$

$$y = \frac{1}{x^{2/3}} = x^{-2/3}$$

$$y' = -\frac{2}{3} x^{-2/3 - 1} = -\frac{2}{3} x^{-5/3} = -\frac{2}{3x^{5/3}}.$$

Example. Differentiate the function $f(t) = \sqrt{t}(a + bt)$

Sln. Using the product rule $g(t) = \sqrt{t}$ $h(t) = a + bt$

$g'(t) = \frac{1}{2\sqrt{t}}$ $h'(t) = b.$

$$f'(t) = g'(t)h(t) + g(t)h'(t) = \frac{1}{2\sqrt{t}} \cdot (a+bt) + \sqrt{t}(b)$$

$$= \frac{a+bt}{2\sqrt{t}} + \frac{b\sqrt{t}}{1} = \frac{a+bt+2b(\sqrt{t})^2}{2\sqrt{t}} = \frac{a+bt+2bt}{2\sqrt{t}}$$

$$= \frac{a+3bt}{2\sqrt{t}}.$$